

TRAINING REPORT ON MILITARY TRAFFICABILITY IN ARID REGION: A REMOTE SENSING APPROACH



**AT THE DEFENCE LABORATORY,
JODHPUR(DRDO)**

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CERTIFICATE

This is to certify that Kartikeya Gaur has undertaken Practical Training on “**Military Trafficability for an Arid Region: A Remote Sensing Approach**”.

The training was conducted at Defence Laboratory, Jodhpur under my supervision and guidance during 06/06/2022 to 30/07/2022.

Date:

Place: Defence Laboratory,
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ABSTRACT-

Military trafficability analysis from a defensive perspective is a step forward. Terrain analysis is important for decision-making framework, with geospatial technologies it can be achieved with other factors also on-board. Topography along with LULC is at its core. The trafficability zone map is produced by effective weighted overlay using DEM derivatives, LULC, and soil strength of the area.

It help identify the obstacles beforehand and is quite useful for planning and distribution of units along the border line. Ultimate aim is to aid satellite data with in-situ field experimentation and other ancillary data, and process it through automated techniques for efficient real-time analysis of the earth surface for better analysis.

KEYWORDS- Remote sensing; off-road trafficability; arid zone; soil strength; terrain analysis.

1. INTRODUCTION-

Trafficability is a measure of how easily vehicles can drive through a particular piece of terrain. Military terrain analysts have a requirement for a tool to conduct trafficability analysis as part of a larger decision-support framework (Sullivan et al., 2003). Trafficability is a very wide and general problem. Units prefer to go through open terrain because it is usually the simplest way. Together with the fact, that soils cover large part of the land the text is aiming only to the trafficability of the soil. There is a lot of critical situations in which the army has to overcome the landscape not only on roads and existent ways. They could be damaged by the enemy or by natural disasters. That is why we have to go through terrain. One of the most important thing, how to solve these situations, is to know the trafficability of soils. The trafficability is the capacity of soils to support vehicles (Cibulova and Sobotkova, 2006). Historically speaking, military has always used geography for its operations, and it is applicable even today. For military geography brings to the fore those physical aspects of concerned locations (combat areas, political boundaries, high tension areas, etc.) which aid in critical decision making for commanders involved. Thus, in military combat areas, it is highly expedient to know and understand the covering terrain and positioning of soldiers is a very important aspect,

.The dominating harsh conditions and undulating terrain make prudent positioning of personnel hugely impactful. The delayed and difficult capturing of Tiger Hill by The Indian Army during the Kargil War demonstrates the role played by position in mountainous terrain (Pundir et al., 2013). Modern geographic information systems (GIS) provide powerful and useful facilities for digitizing and mapping representations of geographic space. They are widely used for performing complex transformations of this data. Layers of a GIS computed with respect to task-relevant properties constitute place vocabularies.

Thus the power of qualitative representations and reasoning can be brought to geospatial applications, riding on the progress made by the commercial world in GIS software. (Donlon & Forbus, 1999)

The land use, slope and soil properties are the key factors to determine the off road trafficability. If the number of factors increased, there are chances that unnecessary information can bias the result and increase the complexity of the problem. The various factors can be evaluated based on their contribution leading to the quantum of resistance offered for the movement of vehicle in off road trafficability scenario (Pundir & Garg, 2019).

A comprehensive evaluation of the contributing factors is very difficult, since between those, there are many related links. (Pundir & Garg, 2020b) GIS is a powerful tool for capturing, storing and analyzing of any type of data. In recent times, the availability of information and advance remote sensing tools, the trafficability can be mapped not only for accessible areas but also for inaccessible areas. It has saved lot of time in comparison of ground survey or in situ measurements and more economical. . (Pundir & Garg, 2022). The research has proven that information about inaccessible terrain can be obtained from satellite/aerial images using GIS and remote sensing with relatively high accuracy. From the result of this research, a field commander can strategies on future operations by knowing which terrain can be accessed

off-road without encountering problem and which terrain are not going to be easily accessible if at all before going into the field. (Sadiya et al., 2017)

The objective of this study is to model accurately and efficiently the different caveats of terrain in a semi-automated mode for improved trafficability of military units and vehicles. The mobilisation of army personnel and heavy machinery plays a crucial role in military tactics, therefore it is important for the top management and the ground-level officers to take a well-thought out decision within the time constraints and this study provides a stimulus for them to take comprehensive decisions based on the different trafficability zones.

2. STUDY AREA-

The study area is located in the south of Jaisalmer district in Indian state of Rajasthan bounded by coordinates of 70°15' 00"E to 70°30'00"E and 26°15'00"N to 26°30'00"N as shown in figure1. Area > 690 km², with a perimeter of more than 105km.

Jaisalmer district is the third largest district of India by area (Census 2011) and it also shares international border with Pakistan. The coarse textured soils occupy 26871 km² (71.0%) area in the district.

Nine soil series spread over four physiographic units viz. dune complex, flat inter-dunal plain, sandy plain with scattered hummocks, and buried pediments have been recognized. Among these, the soils associated with dune complex and some of the flat inter-dunal plain and sandy plain with scattered hummocks are excessively drained, highly sandy and loose with hardly any aggregation. These have poor water retention and nutrient status. (Sharma et al., 2006),

It lies in the Miajalar sub-basin which is structurally simpler. It is characterized by geomorphic forms and landscapes such as dunes, magras, dhands and bhakars. The sand cover is of early Pre-Cambrian gneiss, sedimentary rocks and more recent material deposited by rivers. The surface sand is aeolian (wind deposited) sand of the Quaternary Period.

The desert presents an undulating surface, with high and low sand dunes separated by sandy plains and low, barren hills, or bhakars, which rise from the surrounding plains. Older dunes, are in a semi-stabilized or stabilized condition, and many rise to a height of almost 150 m. Several saline lakes, locally known as dhands, are scattered throughout the region (M. M. Roy and Sharmila Roy, 2019)

In general, the plant species of arid zones have xeromorphological leaf structures, physiological control of transpiration and metabolism, moisture and nutrient storage organs, and thorns (Sharma and Tewari 2016) kumatiyo (*Acacia senegal*), ber (*Zizyphus mauritiana*), and googal (*Commiphora wightii*) have fewer branches, smaller leaves and thorns/spines.

Study Area

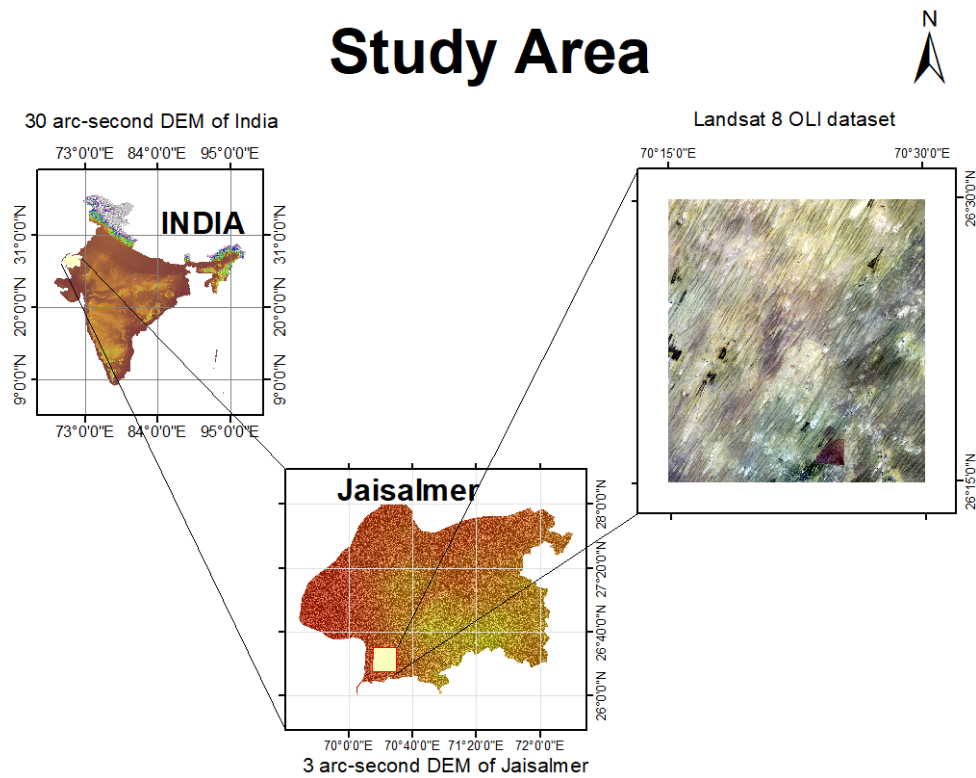


Figure 1: Study Area Jaisalmer.

Soil Type Classification

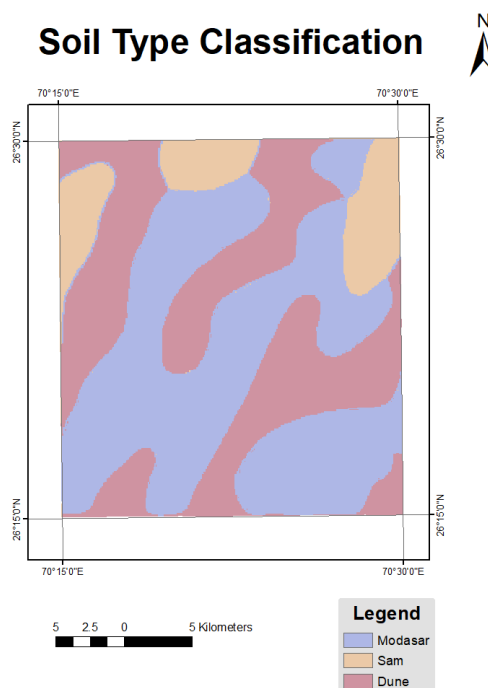


Figure 2: Soil Type Classification.

Table 1 *Textural Classification of soil in the study area.*

S.No.	Physiography	Associated Soil series	Particle Size Distribution (%)				Area	%
			Coarse sand	Fine sand	Silt	Clay		
1	Dune Complex	Dune	10.0	86.2	1.8	2.0	260	38%
2	Flat inter-dunal plain	Modasar	13.3	82.5	1.8	2.4	345	50%
3	Sandy plain with scattered hummocks	Sam	11.7	77.6	4.4	6.3	85	12%
Total							690	100%

3. DATA AND METHODOLOGY-

3.1 Geospatial Data

The resource and data can broadly be grouped into topo-products and instrument based surface data, supplemented by GIS and remote sensing techniques.

The open series maps obtained from were geo-referenced on WGS_84 datum with geographic coordinate system. This information was later projected on Lambert Cylindrical Map Projection (UTM projection, zone 42N) for measuring and analysis.

Multispectral data of Landsat 8 OLI series obtained from USGS and ESA Sentinel-2A obtained from Copernicus were used. The optical bands of both the satellite were enhanced spatial and spectral to produce LULC based on supervised classification with the help LULC thematic layer from ESRI database (Karra et al., 2021) and Google Earth. Thermal bands of Landsat 8 OLI were used to calculate Soil Moisture Index.

1-ARC second SRTM DEM obtained from USGS was used to produce slope map and ruggedness index. Soil Map (Sharma et al., 2006 141) was geo-referenced and converted to vector then to raster for further analysis. The soil data which influence the change in movement of military vehicles were studied from BHUVAN (ISRO-NRSC), NASA SMAP and FAO soil database. The data on moisture condition and soil class present were compiled for the soil strength analysis in the study area.

Table 2 *Datasets Acquired*

S.No.	Satellite/Sensors	Date/Year	Spatial Resolution	Number of bands
1	Landsat 8 OLI/TIRS C2 L1	2022/06/03	30m	11
2	Sentinel-2A	2022/06/24	10m	13
3	SRTM DEM	2000/02/11	1-ARC second	1

Table 3 *Sentinel2A bands*

Band	Resolution	Central Wavelength	Description
B1	60 m	443 nm	Ultra Blue (Coastal and Aerosol)
B2	10 m	490 nm	Blue
B3	10 m	560 nm	Green
B4	10 m	665 nm	Red
B5	20 m	705 nm	Visible and Near Infrared (VNIR)
B6	20 m	740 nm	Visible and Near Infrared (VNIR)
B7	20 m	783 nm	Visible and Near Infrared (VNIR)
B8	10 m	842 nm	Visible and Near Infrared (VNIR)
B8a	20 m	865 nm	Visible and Near Infrared (VNIR)
B9	60 m	940 nm	Short Wave Infrared (SWIR)
B10	60 m	1375 nm	Short Wave Infrared (SWIR)
B11	20 m	1610 nm	Short Wave Infrared (SWIR)
B12	20 m	2190 nm	Short Wave Infrared (SWIR)

Table 4 *Landsat 8 OLI bands*

Bands	Wavelength (micrometers)	Resolution (meters)
Band 1 - Coastal aerosol	0.43-0.45	30
Band 2 - Blue	0.45-0.51	30
Band 3 - Green	0.53-0.59	30
Band 4 - Red	0.64-0.67	30
Band 5 - Near Infrared (NIR)	0.85-0.88	30
Band 6 - SWIR 1	1.57-1.65	30
Band 7 - SWIR 2	2.11-2.29	30
Band 8 - Panchromatic	0.50-0.68	15
Band 9 - Cirrus	1.36-1.38	30
Band 10 - Thermal Infrared (TIRS) 1	10.6-11.19	100
Band 11 - Thermal Infrared (TIRS) 2	11.50-12.51	100

3.2 Methodology and flowchart

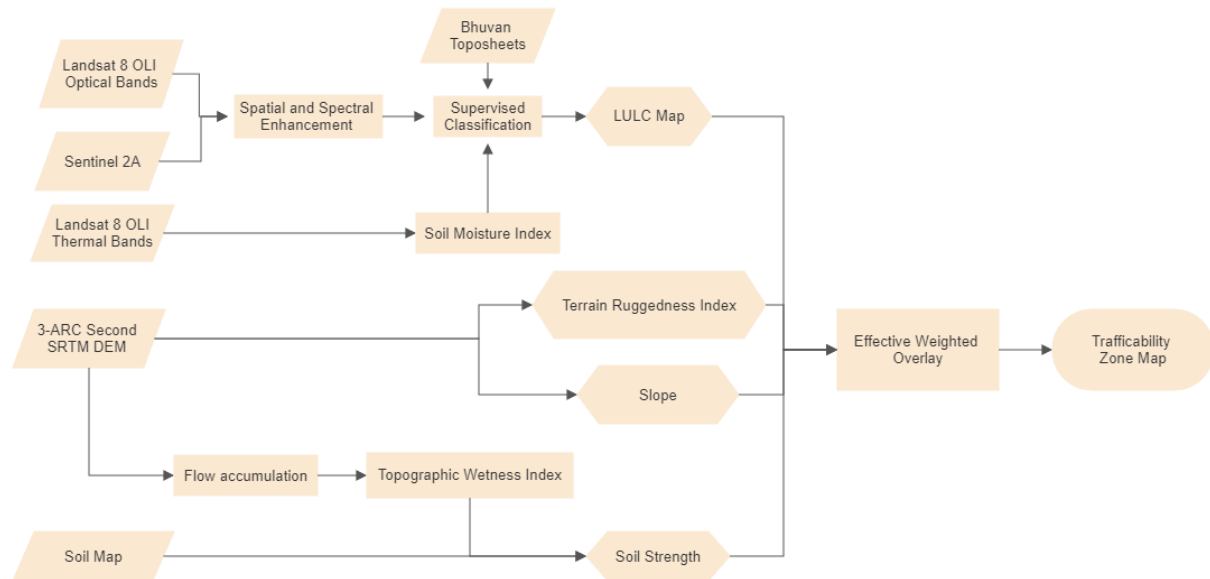


Figure 3 Flowchart portraying purposed methodology

3.2.1 Image Preparation

The subset of the Study area was extracted after creating the band composite of both the satellite datasets.

For Landsat 8 OLI, composite band was produced from bands 1 to 7 (refer Table 4)

For Sentinel2A, composite band was produced from bands 2, 3, 4 & 8 (refer Table 3)

3.2.2 Spatial Enhancement

Pan-sharpening of Landsat data (Figure 4 (b)) was done through the PAN of Sentinel-2A (average of bands 2, 3, 4 & 8).Figure 4 (a).

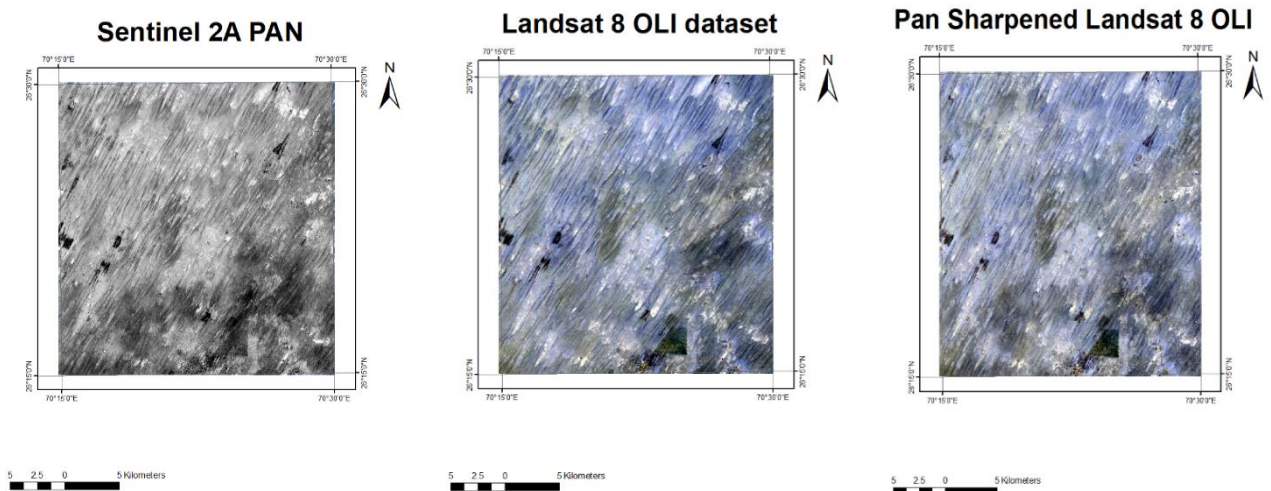


Figure 4(a) Sentinel 2A PAN with 10m resolution

Figure 4 (b) Landsat 8 OLI 7-band composite with 30m resolution

Figure 4(c) Pan-sharpened Landsat 8 OLI dataset with 10m resolution

3.2.3 Spectral Enhancement

Decorrelation stretch was performed and different band ratios and vegetation indices were calculated.

- NDVI (Figure 5(a))
- SAVI(Heute, 1988) (Figure 5(b))
- SMI(Rathore et al., 2018) (Figure 5(c))

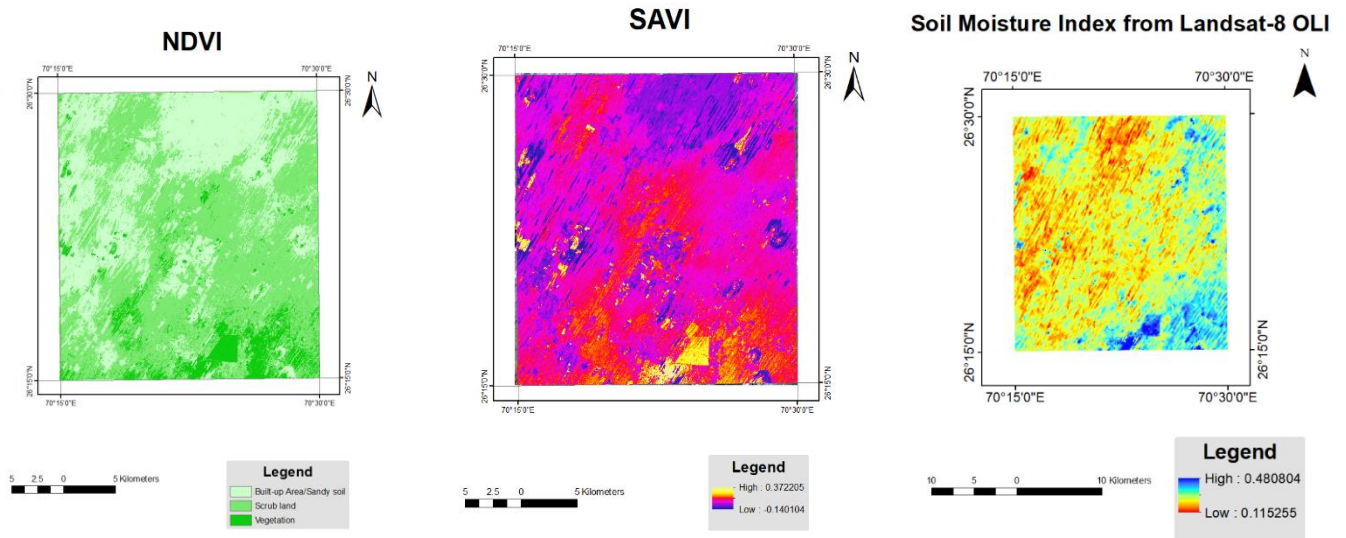


Figure 5(a) Vegetation Map produce through NDVI

Figure 5(b) Soil-adjusted Vegetation Index

Figure 5(c) Soil Moisture Index from Landsat-8 OLI thermal bands

3.2.4 Topographic Analysis

Different derivatives of SRTM DEM was made for surface analysis for better understanding of the terrain and soil moisture.

- Slope
- Hillshade

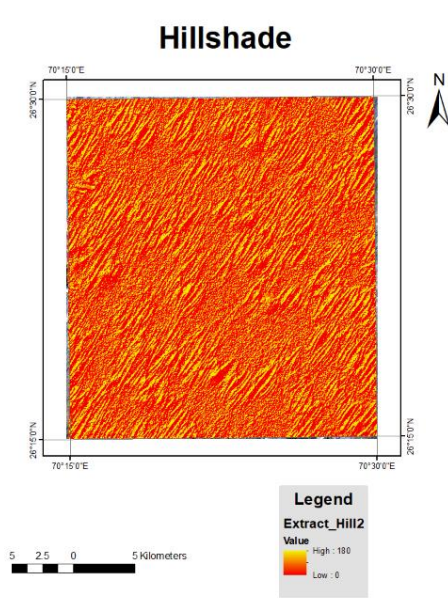


Figure 6(a) Hillahshade derived from SRTM DEM

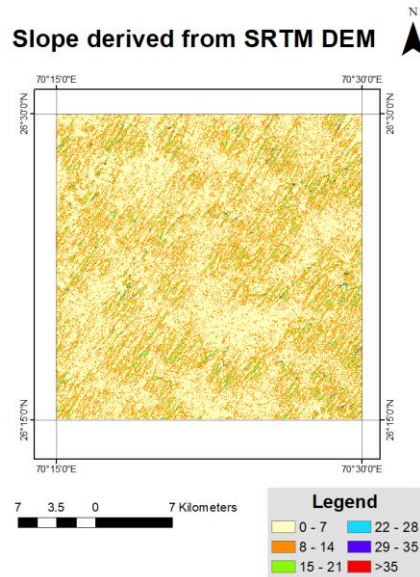


Figure 6(b) Slope derived from SRTM DEM

- c. Terrain Ruggedness Index(Riley et al., 1999)
- d. Topographic Wetness Index(Wild et al., 2021)

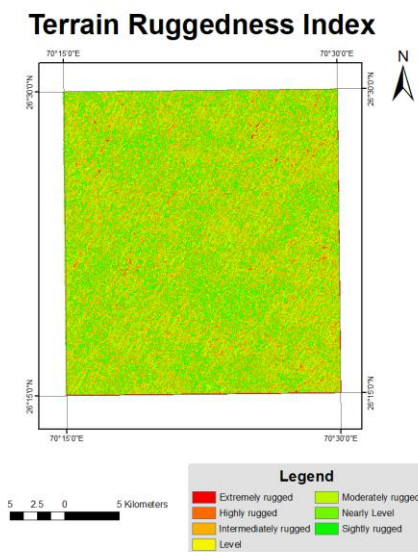


Figure 6(c) Terrain Ruggedness Index

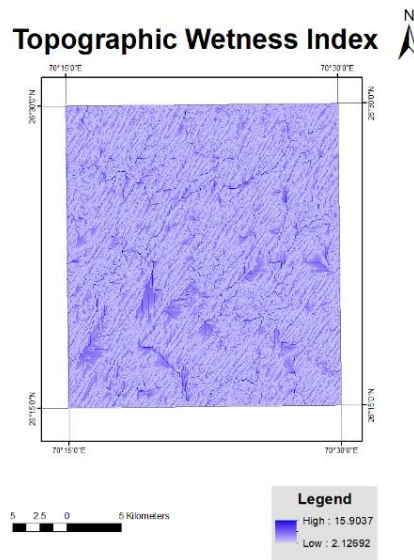


Figure 6(d) Topographic Wetness Index

3.2.5 Interpretation & Feature extraction

Now from the various thematic layer produced different interpretations can be drawn out,

a. Soil Strength Classification

Soil with smaller particles (silt and clay) have a larger surface area than those with larger sand particles, and a large surface area allows a soil to hold more water. A soil with high percentage of silt and clay particle (fine particles), has a higher water-holding capacity. As water content increases, cohesion decreases because increasing water content causes greater separation of clay particles and soften soil cements.

Clay soil has highest water holding capacity and the sand soil has the least, because of particle size of clay particle which are tiny and have many small pore spaces that make water move slower.

From the experimental results, it is found that shear lines under saturated condition pass through very near the point of origin, because sieved soils under saturated condition have little soil cohesion, and that soil which has strong soil cohesion has strong shear strength.(Yokoi, 1968)

Loose sands improve trafficability through an increase in cohesion during rainy periods and return to the loose, less trafficable state during dry periods. (Pundir & Garg, 2022)

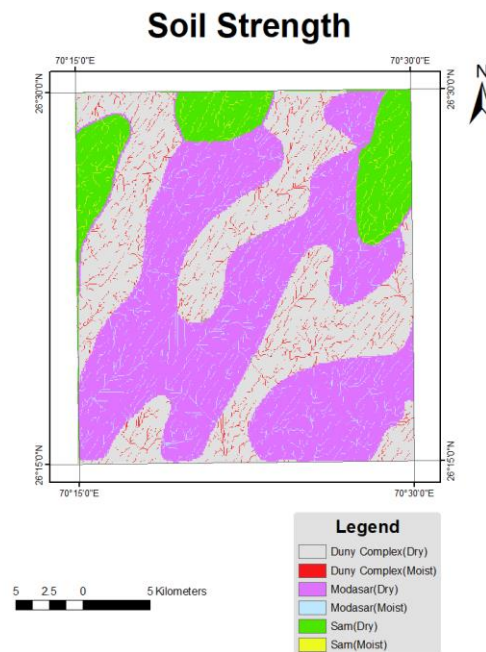


Figure 7 Soil Strength Classification

Table 5 Soil Strength

S.No.	Soil type	Area (km2)	Strength	%
1	Duny Complex(Dry)	246	Moderate	36%
2	Duny Complex(Moist)	14	Weak	2%
3	Modasar(Dry)	326	Strong	47%
4	Modasar(Moist)	19	Moderate	3%
5	Sam(Dry)	81	Weak	12%
6	Sam(Moist)	4	Weakest	1%
Total		690		100%

b. Supervised LULC classification

The pan-sharpened Landsat image was kept as base layer and different thematic layers were used to collect training sets for supervised classification.

Supervised Classification LULC

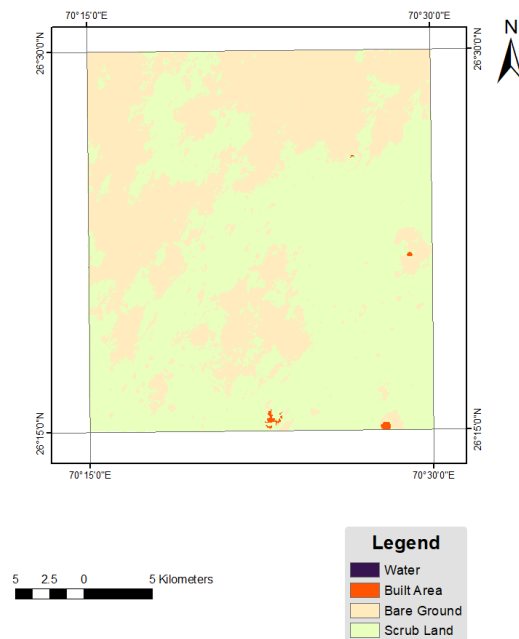


Figure 8 Supervised LULC classification

Table 6 *LULC details of the study area.*

Feature	Area (km2)	%
Bare Ground	271.960	39.4%
Built Area	0.944	0.1%
Scrub Land	417.094	60.4%
Water	0.002	0.0%
Total	690	100%

3.2.6 Data Preparation for Trafficability Zone Map

Effective Weighted overlay was performed according to the table 7, this information is inferred from the different thematic layers.

Table 7 *Weights and rating of different parameters*

Parameters	Rating	Weight
Terrain Ruggedness Index		15%
Extremely rugged	1	
Highly rugged	2	
Moderately rugged	3	
Intermediately rugged	4	
Slightly rugged	5	
Nearly level	8	
Level	9	
Slope		40%
0-7	9	
7-14	7	
14-21	6	
21-28	4	
28-35	3	
>35	1	
LULC		15%
Water	Restricted	
Built-up Area	Restricted	
Bare Ground	9	
Scrub Land	4	
Soil Strength		30%
Duny Complex(Dry)	5	
Duny Complex(Moist)	3	
Modasar(Dry)	9	
Modasar(Moist)	4	
Sam(Dry)	4	
Sam(Moist)	1	

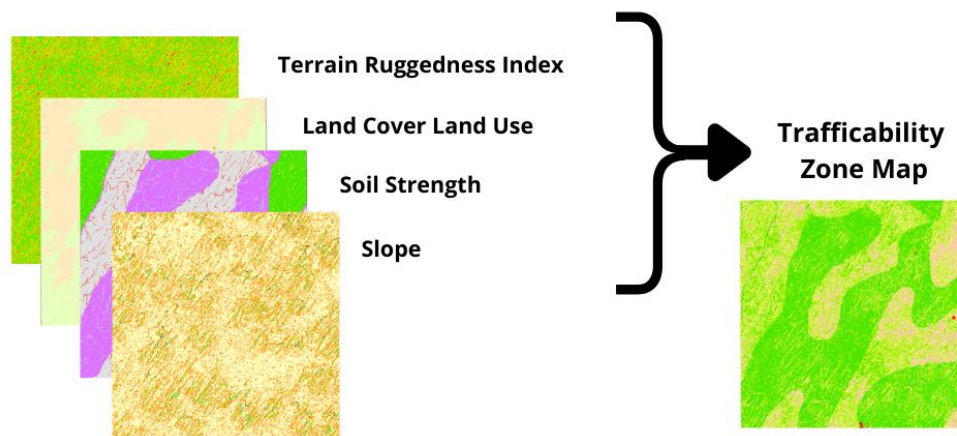


Figure 9. Effective weighted overlay

ESRI ArcGIS 10.3 and ENVI 5.3 softwares and their various tools were used to analyze the datasets.

4. RESULTS AND DISSCUSSIONS-

Figure 8 show the military trafficability map of an arid zone which lies on international border, but trafficability mapping is useful for both military and civil application and can be applied to different terrain conditions. The 690km²(approx.) was divided into 5 trafficability zones.

NO-GO and SLOW-GO represent no trafficability covering 8% and slowest trafficability covering 22% of the study area respectively.

Moderately trafficable can be traversed with low pressure tired vehicle/armored tank/motor cycles and by foot. Meanwhile, GO areas in green colours are locations with no problem of trafficability and it covers 34% of the study area.

S.No.	Trafficability Zone	Area(km ²)	%
1	Go	234	34
2	Moderate Go	246	36
3	No Go	58	8
4	Restricted	1	0
5	Slow Go	150	22
	Total	690	100

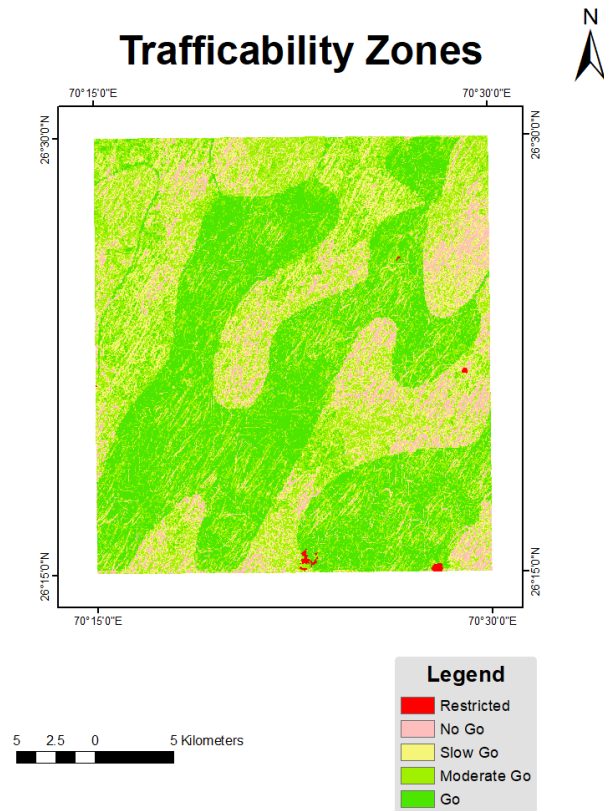


Figure 10 Military Trafficability Zone Map

Percentage wise division of Trafficability Zones

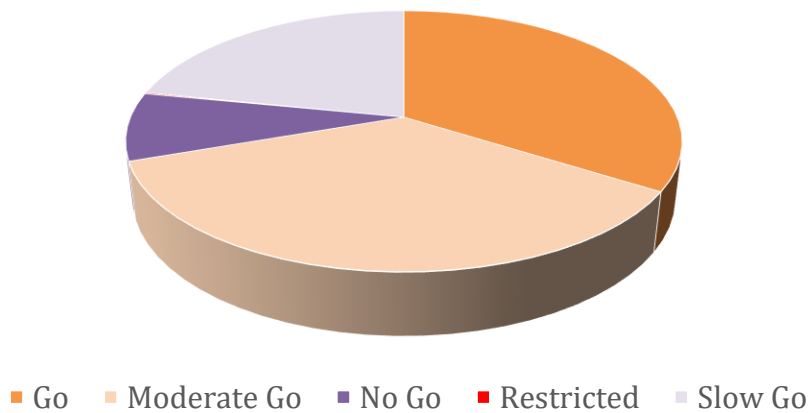


Figure 11 Percentage Area by distribution of different zones

Geospatial analysis from synoptic view sounds feasible and accurate but it has other facets too. The atmospheric condition and cloud cover affect the efficacy of most of the satellite sensors and other similar errors can snowball to inaccurate results. We must always verify the results from various data sources and in-situ analysis ought to be performed in the given resource constraints.

Field survey has its own importance attached, and field test like cone penetrometer and moisture meter help us remove the errors. Due to the variability of soil, trafficability characteristics measured on a particular time cannot be applied later unless some rectification is applied. (Pundir & Garg, 2022) Soil study is complex and there are huge set of parameters apart from particle distribution and soil moisture, which have to be taken into account for further analysis of trafficability.

Trafficability analysis could be vehicle specific also using RCI and VCI, different for wheeled and tracked vehicles (WES, Corps of Engineers, 1963). The analysis could also be carried out with the help of fuzzy logic and machine learning algorithms which will be more accurate and quick.

5. CONCLUSION-

Trafficability analysis mainly depends upon the terrain condition, LULC, water content in the upper most surface and, soil type and strength. These factors are influenced by the climatic condition, landform formations and built-up area. Earlier these parameters were measured by in-situ investigations and cost big quantum of resources, but with advent of remote sensing and GIS technologies and fast-paced advancement in the domain of geospatial analysis provide a viable solution. Military trafficability of an area will aid the ground-level operation and help military, rescue worker in carrying out operations either in war-torn areas or disaster-affected zones. The trafficability zone map will provide a before-hand experience of the affected place which will be a great asset for real-time analysis.

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